Build it, then change it

D103 finance + D117 regression | Editable XLSX | 13 November 2026 | All data are invented classroom assumptions.

1. Rebuild first	Read the module case. On a blank spreadsheet, build the schedules before opening these worked formulas. A populated model is a checking aid, not evidence that you built it.
2. Inspect inputs	Yellow cells are editable assumptions; purple cells are formulas. Values in crore rupees unless a sheet states otherwise. There are no macros, external links or data connections.
3. Run the changes	APV: set year 2 tax-usable flag from 1 to 0. LBO: change exit multiple from 6 to 5. OLS: change only the last development y from 5 to 4. Predict the direction first, then explain the output.
4. Separate fit from use	The future regression batch stays separate from development data. Do not refit on its labels before comparing prediction error. Model arithmetic alone cannot justify a lending or operating decision.
5. Use the checks	Practice is a blank response sheet with revealed baseline targets for self-checking. Use fresh copies for changed cases; baseline targets do not change with model inputs.
What this proves	These formulas were recalculated and tested in LibreOffice. That establishes behavior for stated cases, not Excel/Google Sheets parity, independent learner transfer, current tax law or investment suitability.
Model boundaries	APV: two end-year shields, fair-value debt, stipulated shield risk. LBO: three years, opening-debt interest, positive-EBT cash tax, capped sweep, no losses carried forward. A negative free-cash figure flags a funding gap rather than borrowing automatically.
Keep your work	Save a local copy before editing. No submission, account, certification or saved learner progress is created. Record your reasoning and checks in your own file.

Sources and full written assumptions are in the D103 and D117 module readings. Historical publisher examples are not current market or legal inputs.

D103 | Finite tax shields

All money in crore rupees. Edit yellow cells only. Debt stays outstanding through year 2; shields occur at year end. Flags are 0 or 1.

Unlevered operating value	100.0000
Debt fair value / principal	60.0000
Annual interest rate	10.00%
Stipulated tax rate	25.00%
Shield discount rate	10.00%
Financing-cost PV at time 0	2.0000

Year	Tax usable?	Shield PV
1	1.0000	1.3636
2	1.0000	1.2397

Annual interest	6.0000
Annual potential shield	1.5000
Two-year shield PV	2.6033
Levered operating value	100.6033
Common equity	40.6033
Net financing effect	0.6033

Why not tax rate x debt? The perpetual shortcut does not apply to this two-year schedule. Debt proceeds are not added as free operating value. The financing-cost input is already a present value.

Change E7 from 1 to 0. Explain why APV falls below 100 and why common equity is not APV itself. A different debt or deduction schedule needs a different model, not just a new discount rate.

D103 | Three-year buyout

All money in crore rupees. Synthetic nonfinancial company. No distributions until exit. Cash minimum remains untouched; excess cash after debt payoff accumulates. Model is not an acquisition or borrowing instruction.

Entry enterprise value	80.0000
Fees at closing	2.0000
Minimum funded cash	3.0000
Opening new debt	50.0000
Annual EBITDA	14.0000
Annual depreciation	2.0000
Annual capital spending	3.0000
Annual increase in WC	1.0000
Opening-debt interest rate	10.00%
Positive-EBT cash tax rate	25.00%
Exit EBITDA multiple	6.0000

Entry uses	85.0000
Sponsor equity	35.0000

Line item	Year 1	Year 2	Year 3
Opening debt	50.0000	46.7500	43.2563
Interest	5.0000	4.6750	4.3256
EBT	7.0000	7.3250	7.6744
Cash tax	1.7500	1.8313	1.9186
Net income	5.2500	5.4938	5.7558
Cash before debt sweep	3.2500	3.4938	3.7558
Principal repaid	3.2500	3.4938	3.7558
Closing debt	46.7500	43.2563	39.5005
Unfunded cash shortfall	0.0000	0.0000	0.0000
Excess retained cash	0.0000	0.0000	0.0000
Closing cash balance	3.0000	3.0000	3.0000

Exit EV	84.0000
Equity exit proceeds	47.4995
MOIC	1.3571
Annual IRR (3 years)	10.72%

Cash shortfall row must remain zero for this no-new-borrowing model to be funded. A funding gap returns #N/A for MOIC/IRR. Cash left after full debt repayment accumulates; financing/liquidity choices outside this model are not assumed.

Change only B15 from 6 to 5. The debt schedule must not change. Explain why MOIC drops below 1 even though debt is repaid. IRR formula assumes only time-0 equity and a single year-3 receipt.

D117 | Fit and frozen-batch test

x = load index; y = completion time in hours. Four development rows only. Future labels are not used in the fitted coefficients. Purple values are formulas.

Development x	Observed y	Fitted y	Residual y-yhat	Residual squared
0.0000	1.0000	0.7000	0.3000	0.0900
1.0000	2.0000	1.9000	0.1000	0.0100
2.0000	2.0000	3.1000	-1.1000	1.2100
3.0000	5.0000	4.3000	0.7000	0.4900

x mean	1.5000
y mean	2.5000
Centered x squares	5.0000
Slope	1.2000
Intercept	0.7000
SSE	1.8000
Training MSE (divide n)	0.4500
Residual variance (n-2)	0.9000
SST	9.0000
Training R squared	0.8000

New batch x	Observed y	Frozen fit y	Baseline y	Fit error squared
4.0000	4.0000	5.5000	2.5000	2.2500
5.0000	4.5000	6.7000	2.5000	4.8400

New-batch fit MSE	3.5450
New-batch mean baseline MSE	3.1250

Change B9 from 5 to 4 and refit. Both slope and intercept change. Do not manually edit coefficients. If all development x values are identical, slope is unidentified and the sheet returns #N/A, not a fabricated zero.

Training MSE uses n; residual-variance estimate uses n-2. Neither training R squared nor a two-point future comparison establishes causality, confidence intervals or future population performance. Future x is outside the training range.

Try before revealing the formulas

Blank response cells are for your own baseline calculation. Targets below apply to the original inputs only. Self-check is not grading or a certification.

Baseline task	Your answer	Target	Difference
APV: two-year shield PV		2.6033	
APV: common equity		40.6033	
LBO: entry sponsor equity		35.0000	
LBO: closing debt at year 3		39.5005	
LBO: 6x exit MOIC		1.3571	
OLS: slope		1.2000	
OLS: intercept		0.7000	
OLS: training SSE		1.8000	
OLS: new-batch fit MSE		3.5450	
OLS: new-batch baseline MSE		3.1250	

Changed-case targets: APV with E7=0 gives operating value 99.363636; LBO at 5x gives MOIC 0.957129 and annual IRR -1.449939%; OLS with last development y=4 gives slope 0.9, intercept 0.9 and SSE 0.7.

Explain: why tax rate x debt fails in this APV; why an unfunded LBO cannot be justified by a displayed IRR; why training fit can lose to the mean baseline; why holding future labels out matters. Compare your own reasoning with the module, not just these numeric targets.

Methods: Damodaran APV structure; BIWS debt/cash/exit bridge; Penn State OLS/residual definitions. Publisher links are in the modules. This workbook contains newly authored numbers and formulas, not copied vendor workbooks.